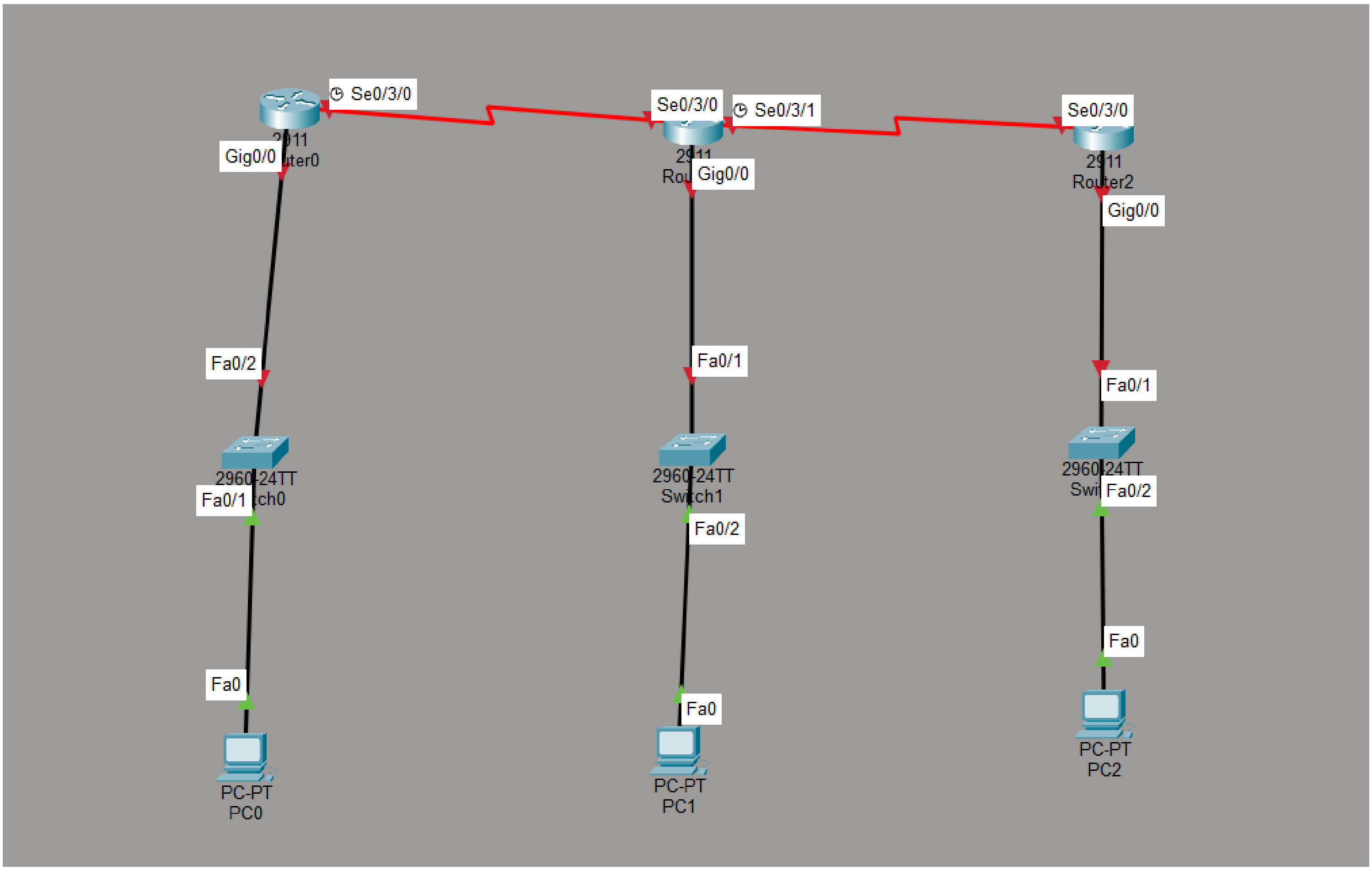
Q-Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

To design and configure a network topology containing three Cisco routers in Cisco Packet Tracer, connect the routers using serial links, connect each router to a LAN, assign IPv4 addresses to all required interfaces and end devices, configure routing, and verify connectivity between the networks.

Network Topology



IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Purpose
Router0	G0/0	192.168.1.1	255.255.255.0	LAN 1 gateway
Router0	S0/3/0	10.0.0.1	255.255.255.252	Link to Router1
Router1	G0/0	192.168.2.1	255.255.255.0	LAN 2 gateway
Router1	S0/3/0	10.0.0.2	255.255.255.252	Link to Router0
Router1	S0/3/1	10.0.0.5	255.255.255.252	Link to Router2

Router2	G0/0	192.168.3.1	255.255.255.0	LAN 3 gateway
Router2	S0/3/0	10.0.0.6	255.255.255.252	Link to Router1
PC0	Fa0	192.168.1.10	255.255.255.0	End device
PC1	Fa0	192.168.2.10	255.255.255.0	End device
PC2	Fa0	192.168.3.10	255.255.255.0	End device

Default gateways:
PC0 → 192.168.1.1
PC1 → 192.168.2.1
PC2 → 192.168.3.1

Router Configuration
Router0

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            unassigned      YES unset  administratively down down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interace serial 0/3/0
                    ^
% Invalid input detected at '^' marker.

Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config)#exit
```

Router 1

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#interface serial 0/3/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.1.0  255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.3.0  255.255.255.0 10.0.0.6
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interaface serial 0/3/0
                        ^
% Invalid input detected at '^' marker.

Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
do wr
Building configuration...
[OK]
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.5
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.5
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verification and Testing

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```